

Context-sensitivity, Pollock's coat, and strictism

Topics in Language and Mind

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1. Pollock's objection to Lewis

(1) If my coat had been stolen last year, it would have been stolen on December 31st.

Tendentious objection: Lewis's theory says (1) is true, but it's false.

A better objection: Lewis's theory says (1) is knowably true, but it's not knowably true.

2. Not just Lewis's problem

Background: a big storm was raging all day on Saturday; it died down late on Saturday night, long after I went to bed. Happily, my favourite tree managed survived the storm.

(2) If this tree had been in pieces on Sunday morning, I would have been too sad to eat breakfast

In the context evoked by (2), it looks like (3) has to be true:

(3) If the tree had been in pieces on Sunday morning, it would have been intact on Saturday at bedtime.

But also (4) and (5) look obviously true:

(4) If the tree had been in pieces on Sunday morning, it would have been destroyed on Saturday.

(5) If the tree had been destroyed on Saturday, it would have been in pieces on Sunday morning.

Now recall the argument-schema CSO ('Weakened Transitivity'):

CSO: $A > B, B > A, A > C \vdash B > C$

Applying CSO to (3)–(5) gives us

(6) If the tree had been destroyed on Saturday, it would have been intact on Saturday at bedtime

This seems bad in the same way as Pollock's coat example. Similarly, applying CSO to (2), (4), (5) yields the similarly problematic

(7) If this tree had been destroyed on Saturday, I would have been too sad to eat breakfast on Sunday

3. Three possible solutions to this problem

- (i) Bite the bullet: say that (6) and (7) are just true, and try to explain away our resistance to them.
- (ii) Deny CSO.
- (iii) Appeal to context-sensitivity. Basic strategy: in the context naturally evoked by (2), all of the above sentences express propositions that we know are true. But in the context naturally evoked by (6) and (7), none of (2), (3), (6), and (7) express propositions that we know to be true.

4. Some support for CSO

We can derive CSO from two other rules whose appeal is more immediately apparent:

RCV ('Very Limited Antecedent Strengthening'): $A > B, A > C \vdash (A \wedge B) > C$

CT ('Cumulative Transitivity'): $A > B, (A \wedge B) > C \vdash A > C$

Here is the derivation:

1.	$A > B$	Premise
2.	$B > A$	Premise
3.	$A > C$	Premise
4.	$(A \wedge B) > C$	1, 3, RCV
5.	$(B \wedge A) > C$	4, Substitutivity
6.	$B > C$	2, 5, CT

We can also derive CSO from RCV together with CA, CC, and ID, thus:

1.	$A > B$	Premise
2.	$B > A$	Premise
3.	$A > C$	Premise
4.	$(A \wedge B) > C$	1, 3, RCV
5.	$(A \wedge B) > (\neg A \vee C)$	4, Weakening
6.	$(\neg A \wedge B) > (\neg A \wedge B)$	ID
7.	$(\neg A \wedge B) > (\neg A \vee C)$	6, Weakening
8.	$((A \wedge B) \vee (\neg A \wedge B)) > (\neg A \vee C)$	7, CA
9.	$B > (\neg A \vee C)$	8, Substitutivity
10.	$B > (A \wedge (\neg A \vee C))$	2, 9, CC
11.	$B > C$	10, Weakening

5. Developing the contextualist response

Basic thought: which closeness and/or accessibility relation is operative in a context depends on which times are relevant in that context. Different antecedents tend to favour

different resolutions of context-sensitivity, by making different times relevant. In the context of (2), Saturday daytime is not relevant, but in the context of (6) and (7), it is.

- An implementation using closeness: match as regards times before the contextually relevant period contributes to closeness; match as regards other times does not.
- An implementation using accessibility: the operative accessibility relation holds between two worlds only when they match as regards history before the contextually relevant time.
 - This could be combined with a strict conditional analysis or the closest-worlds analysis.